

T-test

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1 Student's T-test

William Sealy Gosset is credited with inventing the t-test while working for the Guinness brewery. The idea was refined and supported by the great statistician R. A. Fisher, and the idea was initially described in a paper anonymously by “Student”, in order to protect the commercial interests of Guinness. Today, it is perhaps one of the most prevalent and basic tools in statistics, and [it is a fascinating story](#).

1.1 Objectives

- The question of the t-test
- Data and assumptions
- Graphing
- Test and alternatives
- Practice exercises

2 The question of the t-test

The typical premise of the t-test is that it is used to compare populations you are interested in, which you measure with independent samples. There are a few versions of the basic question.

Compare two independent samples

- Here you have measured a numeric variable and have two samples.
- Are the means of the two samples different (i.e. did the samples come from different populations)?
- Example: An experiment with a control and one treatment group.

Compare 1 sample to a known mean

- Here you have one sample which you wish to compare to a mean value.
- Did the sample come from a population exhibiting the known mean?
- The data are simply a single numeric vector, and the population mean for comparison.

Paired samples

- Here the individual observation comprising the 2 samples are not independent.
- Example: Before vs After experiments
- Another example: Measuring plots that are paired spatially
- For each of these examples, there is a unit, patient, or plot identification, that represents the relationship of each **paired measure**.

3 Data and assumptions

The principle assumptions of the t-test are:

- **Gaussian distribution of observations WITHIN each sample**
- **Heteroscedasticity** (our old friend) - i.e., the variance is equal in each sample
- **Independence of observations**

3.1 Evaluating and testing the assumptions

- The t-test is thought to be somewhat robust to violation of assumptions
- To a certain extent, Gaussian distribution and heteroscedasticity assumption violations won't bias your results.
- The assumption of independence of observations is always of high importance.

Testing assumptions

Gaussian distribution of observations WITHIN each sample

- Plot and evaluate a histogram (`hist()`) and a q-q plot (e.g., with `qqplot()`)
- Use a statistical test evaluating whether data are Gaussian (e.g., `shapiro.test()`)
- NB 1 test EACH SAMPLE SEPARATELY (this is sometimes confusing for beginners).

and If guilty?...

- Mann-Whitney U-test will allow violation of assumptions
- Also called the name Wilcoxon Test

Heteroscedasticity assumption

- Examine graphically

```
# Bartlett's Test
bartlett.test(list(group1, group2))
```

Independence assumption

- Can't really be directly tested without supporting information
- It's up to you to ensure your analysis is being done on the right data
- in time series and spatial data, you can test for autocorrelations (out of scope)

typical Output of t test function.

- The t value; can be positive or negative. The absolute value of t should increase with the probability that the samples came from different populations.
- The degrees of freedom; The number of independent data points available to estimate the t-statistic.
- The 95% confidence interval; This gives a range of values that is likely to contain the true difference in means between the populations from which the samples were taken
- The P-value